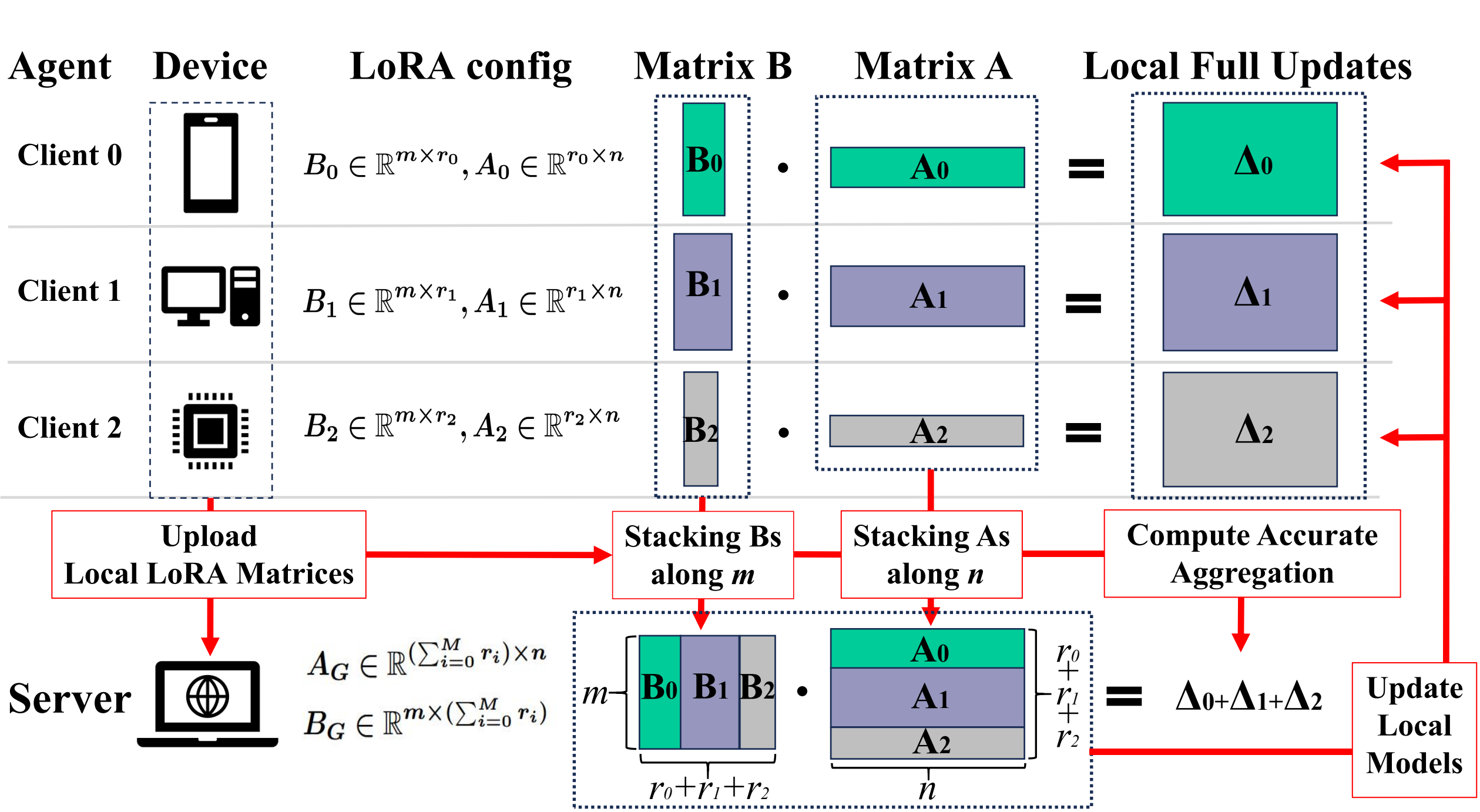
Agent
Device
LoRA config
Matrix B
Matrix A
Local Full Updates
Client 0
$B_0 \in \mathbb{R}^{m \times r_0}, A_0 \in \mathbb{R}^{r_0 \times n}$
B0
A0
Δ0
Client 1
$B_1 \in \mathbb{R}^{m \times r_1}, A_1 \in \mathbb{R}^{r_1 \times n}$
B1
A1
Δ1
Client 2
$B_2 \in \mathbb{R}^{m \times r_2}, A_2 \in \mathbb{R}^{r_2 \times n}$
B2
A2
Δ2
Upload
Local LoRA Matrices
Stacking Bs
along m
Stacking As
along n
Compute Accurate
Aggregation
Server
$A_G \in \mathbb{R}^{(\sum_{i=0}^{M} r_i) \times n}$
$B_G \in \mathbb{R}^{m \times (\sum_{i=0}^{M} r_i)}$
m
B0 B1 B2
A0
A1
A2
r_0
r_1
r_2
$r_0+r_1+r_2$
n
Δ0+Δ1+Δ2
Update
Local
Models